
Honest Inference for Regional Feature Effects

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Abstract

Regional feature-effect methods (e.g., REPID, GADGET, cAME) partition the feature space into subregions where a feature’s effect is more homogeneous and report a per-region effect curve. These curves are reported as point estimates with no statistical inference, leaving practitioners unable to tell whether a regional effect is real or an artifact of where the partition happened to split. We argue this is a post-selection inference problem: the partition is selected from the data, and confidence intervals computed on that same data undercover—severely so in the low signal-to-noise regime that matters most for model auditing. We introduce a method-agnostic *honest* sample-splitting layer that yields valid pointwise and simultaneous confidence bands for the model-conditional regional effect. We prove asymptotic coverage for an *arbitrary* selection rule, and characterize when naive inference fails: severity is governed by the alignment between the selection criterion and the reported effect. Experiments on a unified data-generating process and on REPID’s exact splitting criterion confirm the theory. The method is a drop-in wrapper requiring no change to the underlying partitioner.

1. Introduction

Global feature-effect methods such as partial dependence (PD) plots and accumulated local effects (ALE) summarize the average influence of a feature on a model’s prediction with a single curve, but they can be misleading when the feature interacts with others: averaging over a heterogeneous population hides the very structure an analyst cares about (Molnar et al., 2022). Regional feature-effect methods address this by *partitioning* the feature space into subregions in which the effect of the feature of interest is more homogeneous, and reporting one effect curve per region. REPID

(Herbinger et al., 2022), GADGET (Herbinger et al., 2024), conditional average marginal effects (cAME) (Scholbeck et al., 2024), and the Effector library (Gkolemis et al., 2024) are representative; all fit a tree-like partition by greedily searching splits that maximize the between-region difference (equivalently, minimize within-region heterogeneity) of the effect.

These methods return **point estimates**. A region is reported as “the effect of x_j looks like this here,” with no confidence interval on the curve and no test of whether two regions truly differ. A practitioner therefore cannot answer the most basic question about a regional explanation: *is this regional effect real, or an artifact of where the tree happened to split?* In a model-validation or audit setting—for instance, a credit-risk model reviewed under regulatory guidance such as SR 11-7, where the reliability of an explanation is itself a deliverable—this gap is decisive.

The natural fix, “just add error bars,” is invalid here, and this is the crux of our paper. The partition is *learned from the data*; the tree searches over many candidate features and thresholds for the split that makes the regions look most different. Estimating the regional effect and its confidence interval on the *same* data then double-uses the data: the interval is built around an estimate that was selected precisely because it looked extreme. This is a **post-selection inference** problem, and naive intervals are too narrow. We show the resulting undercoverage can be catastrophic—coverage of a nominal 95% interval falling below 10%—precisely in the weak-signal regime in which an analyst most needs to know whether a discovered region is genuine.

Contributions.

- We identify and formalize the post-selection inference problem in regional feature effects, an issue the literature has not addressed (Section 2).
- We propose a method-agnostic *honest* sample-splitting layer that selects the partition on one part of the data and builds confidence bands on a disjoint part. It yields valid *pointwise* intervals and valid *simultaneous* (studentized sup- t) bands for the model-conditional regional effect, as a drop-in wrapper around any partitioner (Section 2.2).
- We prove asymptotic coverage for an *arbitrary* selec-

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tion rule (Theorem 2.1), and give a $\sqrt{2 \log M}$ characterization of how naive inference fails as the number of competing candidate splits M grows (Proposition 2.2).

- We show empirically that the severity of the problem is governed by the *alignment* between the selection criterion and the reported effect, on a single unified data-generating process and on REPID’s exact splitting criterion; and we delineate a regime (a conservative, non-aligned partitioner) in which the naive interval is comparatively safe (Section 3).

2. Setup and Method

2.1. Regional effects as estimands

We condition throughout on a fixed, trained model $\hat{f} : \mathcal{X} \rightarrow \mathbb{R}$ and target *model-conditional* quantities: uncertainty about what *this* model does, the object relevant for auditing a deployed model. This is the “model effect” rather than the “learner effect” of Molnar et al. (2023); separating out the additional variance due to retraining (the “learner” level, cf. the error decomposition of Heiß et al. (2026)) is out of scope and discussed in Section 5.

Fix a feature of interest s and a per-instance *local-effect functional* $\ell(\hat{f}, \cdot)$ —e.g. a finite-difference derivative $\ell(\hat{f}, x) = (\hat{f}(x + h e_s) - \hat{f}(x - h e_s)) / (2h)$ (ALE-type) or a forward marginal effect (FME/cAME). Write $L_i = \ell(\hat{f}, X_i)$. We observe i.i.d. data $(X_i, L_i)_{i=1}^n$ from a law P (the joint law of covariates and the model-conditional local effect, conditional on $\hat{f}$); L_i has finite variance. For a region $R \subseteq \mathcal{X}$ the (scalar) *regional effect* is

$$\theta(R) = \mathbb{E}[L \mid X \in R] = \frac{\mathbb{E}[L \mathbf{1}\{X \in R\}]}{\mathbb{P}(X \in R)}, \quad (1)$$

with the curve-valued analogue obtained by evaluating an effect curve over a grid of x_s values within R . A region-selection rule $\mathcal{S}$ maps a dataset to a region (the “interesting” leaf of the fitted partition tree). We place **no restriction** on $\mathcal{S}$: it may search greedily over all features and thresholds and overfit arbitrarily. This generality is what makes the honest guarantee below clean and method-agnostic.

2.2. The honest plugin

Naive procedure. Select $\hat{R} = \mathcal{S}(\mathcal{D})$ on the full data $\mathcal{D} = (X_i, L_i)_{i \leq n}$; report the within-region sample mean $\hat{\theta}(\hat{R})$ and the interval $\hat{\theta}(\hat{R}) \pm z_{1-\alpha/2} \hat{\sigma}(\hat{R}) / \sqrt{n_{\hat{R}}}$ from the *same* data. Because $\hat{R}$ and the in-region $\{L_i\}$ are dependent, this interval is not centered at $\theta(\hat{R})$ and its width understates the true dispersion.

Honest procedure (Algorithm 1). Split the indices into disjoint $\mathcal{I}_1$ (selection) and $\mathcal{I}_2$ (inference), *independently of the data values*. Select the region on $\mathcal{D}_1$ only, then estimate

Algorithm 1 Honest regional-effect inference

Input: data $(X_i, L_i)_{i \leq n}$; partitioner $\mathcal{S}$; level α ; bootstrap size B .
 Split indices into disjoint $\mathcal{I}_1, \mathcal{I}_2$ (independent of values).
 $\hat{R} \leftarrow \mathcal{S}((X_i, L_i)_{i \in \mathcal{I}_1})$ {select on $\mathcal{D}_1$ }
 $V \leftarrow \{L_i : i \in \mathcal{I}_2, X_i \in \hat{R}\}$ {infer on $\mathcal{D}_2$ }
 $\hat{\theta} \leftarrow \text{mean}(V)$; bootstrap V (B times)
 Form pointwise band; for curves, form sup- t band
Return: $\hat{R}, \hat{\theta}$, confidence band

the effect and its confidence band on $\mathcal{D}_2$ restricted to the selected region. Because the inference data did not participate in selection, the selected region is—once we condition on $\mathcal{D}_1$ —a fixed, pre-specified set with respect to $\mathcal{D}_2$, and the post-selection target is estimated by a plain i.i.d. sample mean.

Pointwise bands. For a fixed grid point we form a percentile-bootstrap (or Wald) interval from the in-region inference points.

Simultaneous bands. For a curve over a grid $x_s^{(1)}, \dots, x_s^{(m)}$, pointwise intervals do not control coverage of the *whole* curve. We use a studentized sup- t band: from bootstrap draws $\mu_b(\cdot)$ of the regional curve with pointwise standard error $s(\cdot)$, take $T_b = \max_k |\mu_b(k) - \hat{\mu}(k)| / s(k)$, set c to the $(1-\alpha)$ quantile of $\{T_b\}$, and report $\hat{\mu}(k) \pm c s(k)$. This adapts automatically to the effective dimension of the curve (Section 3, Appendix C).

The whole construction is a wrapper: it calls the existing partitioner unchanged and adds an outer split, a bootstrap, and a band.

2.3. Coverage guarantee

Condition on $\hat{f}$. Assume **(A1)** $(X_i, L_i) \stackrel{iid}{\sim} P$ with $\mathbb{E}[L^2] < \infty$; **(A2) honesty:** the index split $(\mathcal{I}_1, \mathcal{I}_2)$ is independent of $\{(X_i, L_i)\}$; **(A3) nondegeneracy:** for the realized region $\mathbb{P}(X \in \hat{R}) > 0$ and $\text{Var}(L \mid X \in \hat{R}) \in (0, \infty)$ (ensured in practice by a minimum-leaf-size constraint).

Theorem 2.1 (Honest coverage, arbitrary selector). *Under (A1)–(A3), conditional on $\mathcal{D}_1$ (so $\hat{R}$ is a fixed set), the studentized honest statistic is asymptotically standard normal as the inference sample grows, and the honest Wald interval satisfies $\mathbb{P}(\theta(\hat{R}) \in \text{CI} \mid \mathcal{D}_1) \rightarrow 1 - \alpha$; marginally, $\mathbb{P}(\theta(\hat{R}) \in \text{CI}) \rightarrow 1 - \alpha$. The result holds for any selection rule $\mathcal{S}$.*

The proof (Appendix A) is short: honesty makes $\hat{R}$ a fixed set with respect to the inference data, so the in-region inference points are an i.i.d. sample from the conditional law $P(\cdot \mid X \in \hat{R})$ and a standard CLT applies; the bootstrap interval is first-order equivalent. Coverage is for the *selected*

region’s effect $\theta(\hat{R})$, the relevant post-selection target.

Proposition 2.2 (Naive failure scales as $\sqrt{2 \log M}$). *Suppose selection chooses, among M candidate regions, the one maximizing $|\hat{\theta}(R_m) - \theta_0|$, where under the null $\theta(R_m) = \theta_0$ for all m and the centered estimates are approximately independent $N(0, s^2)$. Then the selected estimate is displaced from the truth by $\mathbb{E}|\hat{\theta}(\hat{R}) - \theta_0| = s \cdot \Theta(\sqrt{2 \log M})$ while the naive half-width stays $z_{1-\alpha/2} s$; hence naive coverage $\rightarrow 0$ as $M \rightarrow \infty$, whereas honest coverage is unaffected.*

Proposition 2.2 (proof and Monte-Carlo check in Appendix A) explains the mechanism: undercoverage is driven by the number of competing candidates the selector exploits, and vanishes once a strong true signal makes $\hat{R}$ effectively deterministic.

3. Experiments

Unified setup. All experiments use one data-generating process with a single signal knob b and a single noise knob σ . The model-conditional ICE curve of the feature of interest x_1 for instance i is $\text{ICE}_i(x_1) = b x_{2,i} x_1 + \varepsilon_i(x_1)$ with $\varepsilon \sim N(0, \sigma^2)$ i.i.d. over instances and grid points; the deterministic part is a clean bilinear interaction (local derivative $b x_2$), and ε is the estimation noise of the local effect that the selector can overfit. Feature x_1 is the feature of interest, x_2 the single true effect-modifier, and the remaining columns ($d = 16$ in total) are pure noise. With $\sigma = 1$ fixed, b equals the signal-to-noise ratio b/σ , giving the scalar and curve experiments one comparable axis. The truth is the model-conditional regional effect, $\theta(R) = b \mathbb{E}[x_2 | X \in R]$ (scalar) or its linear curve analogue, computed on a large independent oracle sample. Selectors are *aligned* with the reported effect: greedy variance-reduction of the local effect (scalar) and REPID’s exact splitting criterion (curve)¹. We report coverage of nominal 95% intervals with Monte-Carlo standard errors over R replications. Full DGP equations, hyperparameters, and tables are in Appendix B.

3.1. The disease and the cure

Figure 1 summarizes the central result. Panel (a): for the scalar regional effect, naive coverage collapses to **0.085** at the null ($b/\sigma = 0$) and climbs back toward nominal as the signal grows, while honest coverage stays at ≈ 0.95 throughout. The overlaid probability that the split lands on a pure-noise feature tracks the collapse almost exactly, identifying the winner’s curse as the mechanism. Panel (b): for the curve (REPID criterion), naive pointwise coverage falls to **0.78** at the null and recovers, while honest stays at

¹REPID minimizes the within-region L_2 variance of mean-centered ICE curves (Herbinger et al., 2022); we reimplement it exactly.

Table 1. Coverage of nominal 95% intervals on the unified DGP (model-conditional truth; MC SE in parentheses). Naive undercovers severely at low signal-to-noise; honest is at nominal throughout.

b/σ	SCALAR		CURVE (POINTWISE)	
	NAIVE	HONEST	NAIVE	HONEST
0.0	0.085	0.965	0.777	0.948
0.3	0.875	0.960	0.931	0.939
0.6	0.925	0.955	0.938	0.949
1.0	0.932	0.953	0.944	0.949
2.0	0.927	0.955	0.942	0.945

nominal. Headline numbers are in Table 1.

The reading is uniform: the post-selection problem is *severe at low signal-to-noise*—the regime in which an analyst cannot already see the structure and most needs a valid test—and *negligible at high signal-to-noise*, where the selector reliably locks onto the true split. Honest inference is valid across all regimes, at a width cost of roughly $\sqrt{2}$ (e.g. scalar honest width 0.38 vs. naive 0.26), the price of estimating on half the data.

3.2. Simultaneous bands

Figure 1(c) isolates the curve-specific multiplicity issue. Because the per-grid-point estimation noise decorrelates the curve, the honest *pointwise* band has simultaneous coverage of only ≈ 0.52 – 0.62 : it covers each point at nominal rate but leaves *some* point uncovered most of the time. The studentized *sup-t* band restores simultaneous coverage to ≈ 0.92 – 0.96 at a modest width increase. The *sup-t* band is a safe default: it never undercovers and, when the curve’s variation is low-rank (a near-linear or monotone regional effect), it adapts automatically without meaningful widening (Appendix C).

3.3. When does it matter? Alignment and a real library

The disease is severe under an *aligned* selector—one whose criterion optimizes the same effect statistic that is then reported—as demonstrated above on REPID’s exact criterion. This alignment is not automatic. The default `RegionalALE` of `Effector` (Gkolemis et al., 2024) selects by *heterogeneity reduction* rather than by maximizing the regional mean, and is comparatively conservative; in our experiments it either recovers the true split or declines to split, rarely chasing noise, so naive coverage of the model-conditional effect is close to nominal. The practical lesson is not that honesty is always necessary, but that one cannot know the regime in advance, so honest splitting is a cheap insurance: negligible cost when the selector is conservative, and protection against catastrophic undercoverage when it is aligned and the signal is weak. Finally, honesty addresses post-selection bias but not model bias: if one instead targets

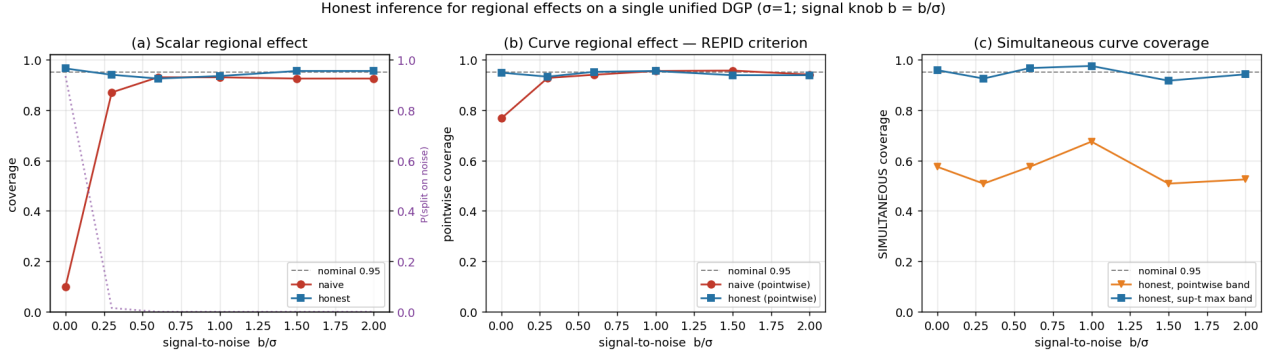


Figure 1. Honest inference for regional effects on a single unified DGP ($\sigma = 1$, so the signal knob b equals b/σ). (a) Scalar regional effect: naive coverage (red) collapses to 0.085 at the null and recovers; honest (blue) is flat at nominal; the dotted purple curve is the probability the split lands on a noise feature. (b) Curve regional effect under REPID’s exact criterion: naive pointwise coverage falls to 0.78 at the null; honest is flat. (c) Simultaneous curve coverage: the honest *pointwise* band (orange) covers the whole curve only ≈ 0.55 of the time, while the studentized sup- t band (blue) restores ≈ 0.95 . Nominal 0.95 dashed.

the data-generating effect with a refit model, honest intervals remain centered correctly for the model-conditional target but the model’s own bias is a separate, locatable term (Appendix D).

4. Related Work

Regional effects. REPID (Herbinger et al., 2022), GADGET (Herbinger et al., 2024), cAME (Scholbeck et al., 2024), and Effector (Gkolemis et al., 2024) fit interpretable regional effect curves; GADGET’s PINT is an interaction *detection* procedure—its authors note it is not a formal test—and does not provide uncertainty on the regional curves. Conditional inference trees (Hothorn et al., 2006), used inside cAME, test association at a split but do not produce post-split curve intervals.

Uncertainty for feature effects. Okoli (2023) adds bootstrap confidence intervals for *global* ALE; Moosbauer et al. (2021) pair partitioning with confidence bands but for a Gaussian-process surrogate, not model-agnostically; Molnar et al. (2023) and Heiß et al. (2026) analyze global PD/ALE as statistical estimators and decompose their error. All operate at the global level and none addresses the post-selection problem at the *regional* level.

Honest estimation. Sample splitting for valid post-selection inference goes back to Cox (1975) and Wasserman & Roeder (2009); honesty for data-selected partitions is due to Athey & Imbens (2016) and Wager & Athey (2018), whose base central limit theorem is for conditional means and is non-causal, so the subgroup-mean guarantee exists in spirit. Our contribution is to recognize that regional feature-effect methods incur exactly this selection problem, to port the honest guarantee to the regional-effect estimand as a

method-agnostic plugin, and to characterize when it bites (selection–inference alignment). Selective inference without sample splitting (Lee et al., 2016) is a more efficient but harder alternative, left to future work.

5. Discussion and Conclusion

We have shown that regional feature-effect methods incur a post-selection inference problem that makes naive confidence intervals undercover—severely in the weak-signal regime that matters most for auditing—and that an honest sample-splitting wrapper restores valid pointwise and simultaneous coverage for an arbitrary selector, at a modest width cost.

Limitations. The guarantee is model-conditional; learner-level inference would additionally require model-variance terms, locatable within the decomposition of Heiß et al. (2026). Sample splitting costs roughly $\sqrt{2}$ in interval width; cross-fitting is not a free remedy here because the two folds generally select *different* regions and thus estimate different quantities. Valid non-split (selective) inference (Lee et al., 2016) is the natural next step.

Conclusion. Honest splitting is cheap insurance for regional explanations: validity matters exactly where auditing is hardest—when the signal is weak and the partition is selected by optimizing the reported effect.

Impact Statement

This paper aims to make explanations of machine-learning models more statistically reliable, which supports responsible model auditing and regulatory review. We see no specific negative societal impacts beyond those already attendant to

interpretable machine learning.

Software and Data

All experiments use synthetic data; code to reproduce every figure and table will be released. Generative AI tools were used to assist with drafting and code scaffolding; the authors take full responsibility for the content.

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A. Theory: full statements and proofs

We restate the setup for self-containedness. Condition on the fixed model $\hat{f}$. Data $(X_i, L_i)_{i=1}^n$ are i.i.d. from P with $\mathbb{E}[L^2] < \infty$ (A1). The honest split $(\mathcal{I}_1, \mathcal{I}_2)$ is independent of the data values (A2). For the realized region $p_{\hat{R}} := \mathbb{P}(X \in \hat{R}) > 0$ and $v_{\hat{R}} := \text{Var}(L \mid X \in \hat{R}) \in (0, \infty)$ (A3). The estimand is $\theta(R) = \mathbb{E}[L \mid X \in R]$. On the inference sample define $n_{2,R} = \#\{i \in \mathcal{I}_2 : X_i \in R\}$, $\hat{\theta}_2(R) = n_{2,R}^{-1} \sum_{i \in \mathcal{I}_2 : X_i \in R} L_i$, and $\hat{v}_2(R)$ the within-region sample variance.

Theorem A.1 (restatement of Theorem 2.1). *Under (A1)–(A3), conditional on $\mathcal{D}_1$ (so $\hat{R} = R$ is a fixed set), as $n_2 \rightarrow \infty$,*

$$\sqrt{n_{2,\hat{R}}} \frac{\hat{\theta}_2(\hat{R}) - \theta(\hat{R})}{\sqrt{\hat{v}_2(\hat{R})}} \xrightarrow{d} N(0, 1),$$

hence $\mathbb{P}(\theta(\hat{R}) \in [\hat{\theta}_2(\hat{R}) \pm z_{1-\alpha/2} \sqrt{\hat{v}_2(\hat{R})/n_{2,\hat{R}}}] \mid \mathcal{D}_1) \rightarrow 1 - \alpha$, and marginally $\rightarrow 1 - \alpha$. This holds for any selection rule $\mathcal{S}$.

Proof. Step 1 (honesty $\Rightarrow$ fixed region). Condition on $\mathcal{D}_1$. By (A2) the inference rows $\{(X_i, L_i) : i \in \mathcal{I}_2\}$ are independent of $\mathcal{D}_1$ and i.i.d. $\sim P$. Since $\hat{R} = \mathcal{S}(\mathcal{D}_1)$ is determined by $\mathcal{D}_1$ alone, once we condition on $\mathcal{D}_1$ the region $\hat{R} = R$ is a deterministic, fixed set, independent of $\mathcal{D}_2$. This is the crux: honesty turns the post-selection target into a pre-specified one.

Step 2 (in-region points are i.i.d. from the conditional law). For each $i \in \mathcal{I}_2$, independently, $\mathbb{P}(X_i \in R) = p_R$, and given $X_i \in R$, $(X_i, L_i) \sim P(\cdot \mid X \in R)$. Hence $n_{2,R} \sim \text{Binomial}(n_2, p_R)$ and, conditional on the in-region index set, the corresponding L_i are i.i.d. with mean $\theta(R)$ and variance v_R .

Step 3 (CLT + Slutsky for fixed R). Conditional on $\mathcal{D}_1$ and $n_{2,R} = k$, $\hat{\theta}_2(R)$ is an average of k i.i.d. mean- $\theta(R)$, variance- v_R variables; by the Lindeberg–Lévy CLT (finite variance, A3), $\sqrt{k}(\hat{\theta}_2(R) - \theta(R)) \xrightarrow{d} N(0, v_R)$ while $\hat{v}_2(R) \xrightarrow{p} v_R$ by the LLN. Slutsky gives the studentized statistic $\rightarrow N(0, 1)$.

Step 4 (random count). By the LLN $n_{2,R}/n_2 \xrightarrow{a.s.} p_R > 0$, so $n_{2,R} \rightarrow \infty$ a.s.; the limit in Step 3 is the same for every diverging k , so an Anscombe random-index argument yields the convergence with the random count. No uniformity over R is needed because, conditional on $\mathcal{D}_1$, R is fixed.

Step 5 (coverage). The studentized convergence gives conditional coverage $\rightarrow 1 - \alpha$ for the fixed target $\theta(\hat{R})$. By the tower property, $\mathbb{P}(\theta(\hat{R}) \in \text{CI}) = \mathbb{E}[\mathbb{P}(\theta(\hat{R}) \in \text{CI} \mid \mathcal{D}_1)] \rightarrow 1 - \alpha$. $\square$

Remark A.2 (bootstrap). The percentile-bootstrap interval used in the implementation resamples the in-region inference points and is first-order equivalent to the Wald interval for a sample mean; under (A1)–(A3) it inherits the same coverage guarantee.

Remark A.3 (curve version and simultaneous bands). For a curve over a grid, applying the theorem at each grid point gives pointwise coverage. Joint coverage needs a multiplicity adjustment; the studentized sup- t band of Section 2.2 provides it and adapts to the effective dimension of the curve (Appendix C).

Proposition A.4 (restatement of Proposition 2.2). *Suppose selection chooses, among M candidate regions $R_1, \dots, R_M$, the one maximizing $|\hat{\theta}(R_m) - \theta_0|$, where (null) $\theta(R_m) = \theta_0$ for all m and the centered estimates are approximately independent $N(0, s^2)$. Then $\mathbb{E}|\hat{\theta}(\hat{R}) - \theta_0| = s \cdot \Theta(\sqrt{2 \log M})$, while the naive half-width stays $z_{1-\alpha/2}s$; the standardized distance of the truth from the naive interval grows like $\sqrt{2 \log M}$, so naive coverage $\rightarrow 0$.*

Sketch. The maximum of M i.i.d. half-normals concentrates at order $s\sqrt{2 \log M}$ by extreme-value asymptotics; selection centers $\hat{\theta}(\hat{R})$ that far from θ_0 while leaving the within-region standard error unchanged, so the truth falls an $\Omega(\sqrt{2 \log M})$ multiple of half-widths outside the interval. The exact conditional law of $\hat{\theta}(\hat{R})$ is the right-skewed law of the selected maximum, not a shifted normal, so we claim only the scaling and the collapse, not a closed-form coverage value. $\square$

Numerical confirmation. A self-contained Monte-Carlo check (pure null, $L_i \sim N(0, 1)$, $\theta(R) = 0$; selection = arg max over M noise-feature splits of $|\hat{\theta}|$; honest = 50/50 split) gives, at nominal 0.95:

n	M	naive coverage	honest coverage
1000	20	0.698	0.943
1000	50	0.609	0.954

Honest hits nominal for the selected region as Theorem 2.1 predicts; naive undercovers and worsens with M as Proposition 2.2 predicts.

B. Unified DGP: details and full tables

Model. The model-conditional ICE curve of x_1 for instance i is

$$\text{ICE}_i(x_1) = b x_{2,i} x_1 + \varepsilon_i(x_1), \quad \varepsilon \sim N(0, \sigma^2) \text{ i.i.d. over instances and grid points.}$$

The deterministic part is $f(x) = b x_2 x_1$ (local derivative wrt x_1 equal to $b x_2$); ε is the estimation noise of the local effect. Features: x_1 (col. 0, feature of interest, never a split candidate), x_2 (col. 1, the one true modifier), columns 2, $\dots$, $d-1$ pure noise; all Uniform $[-1, 1]$, $d = 16$. Signal knob b ($x_2 \rightarrow \text{slope}$), noise knob σ ; SNR = b/σ . The scalar local effect is $L_i = b x_{2,i} + \varepsilon_i$. Truths (model-conditional, population): scalar $\theta(R) = b \mathbb{E}[x_2 \mid R]$; curve $b \mathbb{E}[x_2 \mid R] (x_1^{(k)} - \bar{x}_1)$, computed on an oracle sample of 4×10^5 points.

Full coverage tables. Scalar (Wald CI; $n = 1000$, $R = 400$ replications; MC SE in parentheses):

b/σ	naive cov	honest cov	naive width	honest width	$\mathbb{P}(\text{split on noise})$
0.0	0.085 (.014)	0.965 (.009)	0.258	0.376	0.93
0.3	0.875 (.017)	0.960 (.010)	0.209	0.315	0.01
0.6	0.925 (.013)	0.955 (.010)	0.194	0.284	0.00
1.0	0.932 (.013)	0.953 (.011)	0.193	0.279	0.00
2.0	0.927 (.013)	0.955 (.010)	0.206	0.293	0.00

Curve (REPID exact criterion; bootstrap bands; $n = 1500$, $R = 300$):

b/σ	naive pointwise	honest pointwise	naive sup- t simult.	honest sup- t simult.
0.0	0.777 (.024)	0.948 (.013)	0.470	0.957
0.3	0.931 (.015)	0.939 (.014)	0.887	0.920
0.6	0.938 (.014)	0.949 (.013)	0.927	0.933
1.0	0.944 (.013)	0.949 (.013)	0.937	0.960
2.0	0.942 (.014)	0.945 (.013)	0.940	0.930

C. Studentized sup- t band: when multiplicity matters

From bootstrap draws $\mu_b(\cdot)$ of the regional curve with pointwise SE $s(\cdot)$, set $T_b = \max_k |\mu_b(k) - \hat{\mu}(k)|/s(k)$, $c =$ the $(1 - \alpha)$ quantile of $\{T_b\}$, band $\hat{\mu}(k) \pm c s(k)$. The band adapts to the effective dimension of the curve: when the bootstrap variation is low-rank (near-linear/monotone regional effect, dominated by the regional slope), the grid points are highly correlated, pointwise and simultaneous coverage nearly coincide, and the band barely widens; when the variation is multi-dimensional (per-grid-point noise decorrelates the curve), the pointwise band's simultaneous coverage falls well below nominal and the sup- t band restores it. Honest arm, nominal 0.95:

regime	band	pointwise cov	simultaneous cov	width
low-rank curve	pointwise	0.963	0.963	0.152
low-rank curve	sup- t	0.963	0.963	0.153
decorrelated curve	pointwise	0.953	0.787	0.119
decorrelated curve	sup- t	0.997	0.975	0.157

D. Alignment and a real library (Effector)

The severity of post-selection undercoverage depends on whether the selection criterion is *aligned* with the reported statistic. Our scalar greedy selector and REPID's exact criterion both directly optimize a within-region effect quantity and are aligned; they exhibit the severe undercoverage of Section 3.

Effector's default RegionalALE (Gkolemis et al., 2024) instead selects by heterogeneity reduction and is comparatively conservative. On a deterministic analytic model it reliably recovers the true split or declines to split, rarely chasing noise;

under model-conditional truth, naive and honest coverage are both near nominal. This is the boundary case in which the naive interval is comparatively safe—but it cannot be assumed in advance, which is why honest splitting is recommended as a default.

Model- vs. DGP-conditional truth. If one instead targets the *data-generating* regional effect using a refit model, two effects compound. In that setting we observe naive coverage near 0 and honest coverage also well below nominal, because honesty removes the post-selection bias but not the model’s own bias relative to the DGP; the latter is a separate term, locatable in the decomposition of [Heiß et al. \(2026\)](#). All coverage claims in the main text are model-conditional, the relevant target for auditing a deployed model.